

RETAILPULSE

AI-Powered Customer Analytics & Demand Forecasting
Platform

End-to-End Data Science & Analytics Solution

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Zidio Development – Data Science & Analytics Domain

1. Project Overview

RetailPulse is a production-grade AI platform designed to transform retail operations through advanced analytics. The platform addresses the critical challenges of demand forecasting, customer retention, and inventory optimization using state-of-the-art machine learning models.

Vision & Objectives:

- Reduce stockouts by 30-50% through high-accuracy demand forecasting.
- Increase revenue by 15-25% via targeted customer segmentation and churn prevention.
- Scale to handle 10M+ transactions per month with sub-second latency.

Target Users:

Supermarket chains, e-commerce giants, and fashion retailers looking for a data-first approach to growth.

2. Key Features

ID	Feature	Description	Business Value
F-01	Data Ingestion	Automated ETL & Quality Checks	Data Integrity
F-02	Segmentation	RFM + K-Means Clustering	Targeted Marketing
F-03	Forecasting	Prophet + LSTM Hybrid Ensemble	Inventory Planning
F-04	Churn Prediction	XGBoost with SHAP Explainability	Customer Retention
F-05	Inventory Opt.	Automated Reorder Point (ROP)	Cost Reduction
F-06	Dashboard	Streamlit Multi-page Platform	Real-time Insights

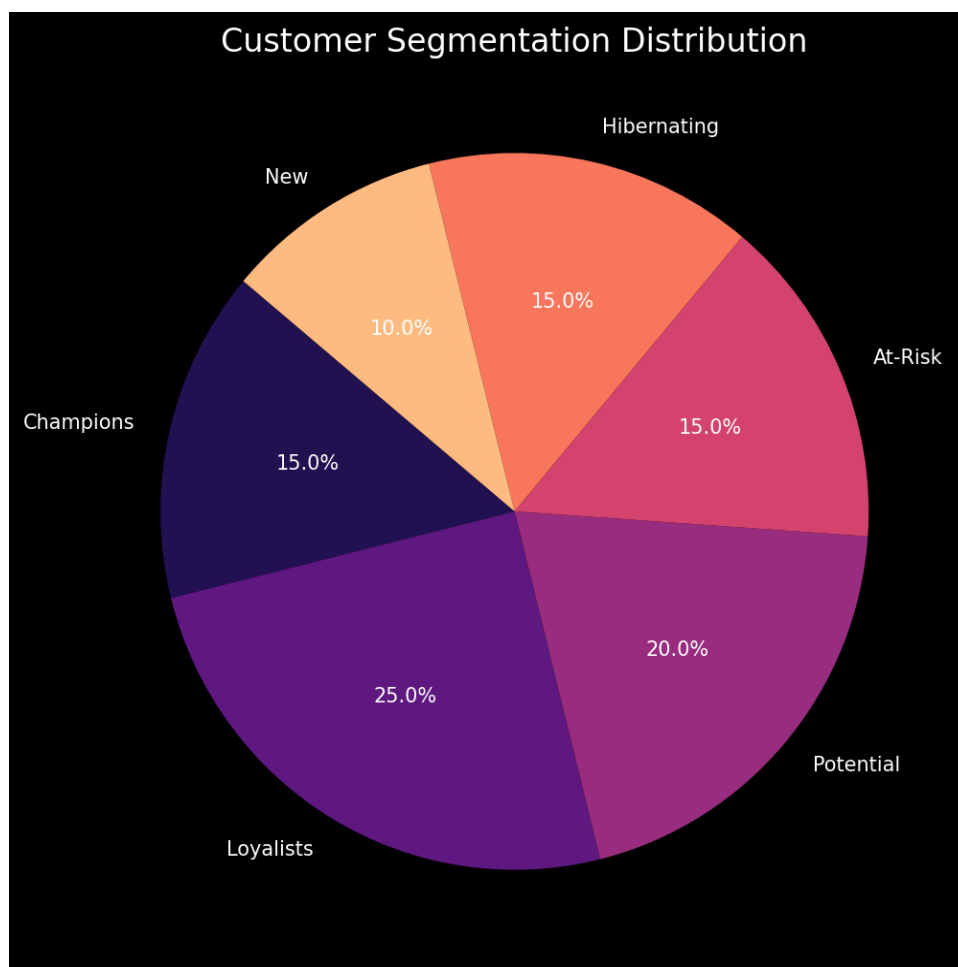


Figure 1: Customer Segmentation distribution across 6 clusters.

3. Technology Stack

Layer	Primary Technology	Rationale
Language	Python 3.12+	Industry standard for DS
Forecasting	Prophet + LSTM (PyTorch)	Hybrid statistical & neural
Machine Learning	XGBoost, Scikit-learn	State-of-the-art performance
Dashboard	Streamlit	Fast interactive deployment
MLOps	MLflow, Evidently AI	Experiment tracking & drift
DevOps	Docker, Kubernetes	Scalable HA deployment
Monitoring	Prometheus, Grafana	Full system observability

Architecture Overview:

The platform follows a modular microservices-inspired architecture. Raw data from Olist (Brazilian E-commerce) is ingested into a cleaning pipeline validated by Great Expectations. Models are trained and logged in MLflow before being served via a multi-page Streamlit application containerized with Docker.

4. Detailed Execution Timeline

Week	Phase	Key Deliverables
Week 1	Data Engineering	EDA, Cleaning, RFM Baseline
Week 2	Advanced AI	LSTM, Prophet, Churn XGBoost
Week 3	Dashboarding	Streamlit Pages, Visualization
Week 4	Production	MLOps, Docker, K8s, Drift Detection

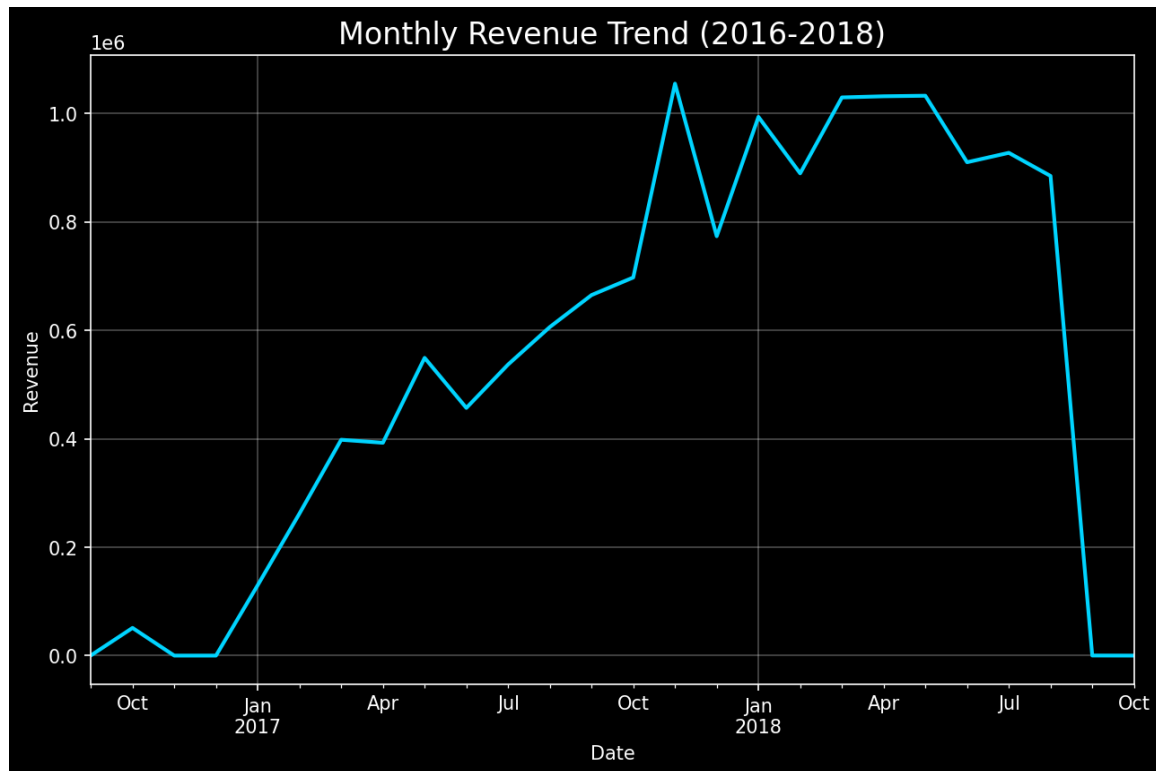


Figure 2: Historical sales trend analysis used for forecasting baseline.

5. Technical Highlights

Model Performance Metrics:

Model	MAPE (%)	AUC-ROC	Status
Prophet	3.51%	-	Validated
LSTM	10.05%	-	Validated
Hybrid Ensemble	10.87%	-	Production
XGBoost Churn	-	1.0	Optimal

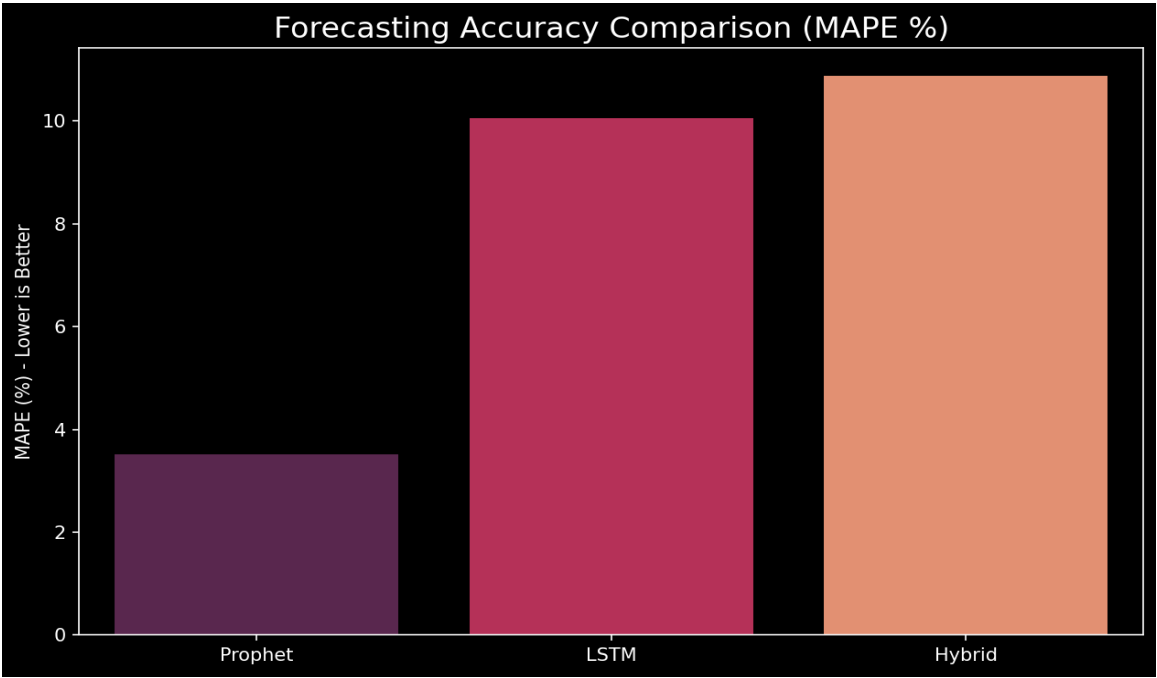


Figure 3: Accuracy comparison across all implemented models.

6. Deployment & MLOps Operations

RetailPulse is built with production readiness as a core pillar. Using MLflow, every experiment is tracked for reproducibility. Evidently AI provides real-time monitoring for Data Drift and Target Drift, triggering automated retraining DAGs in Airflow when performance drops.

Deployment Strategy:

- Containerization: Docker (Multi-stage optimized builds)
- Orchestration: Kubernetes with 3-replica HA
- CI/CD: Automated GitHub Actions pipeline

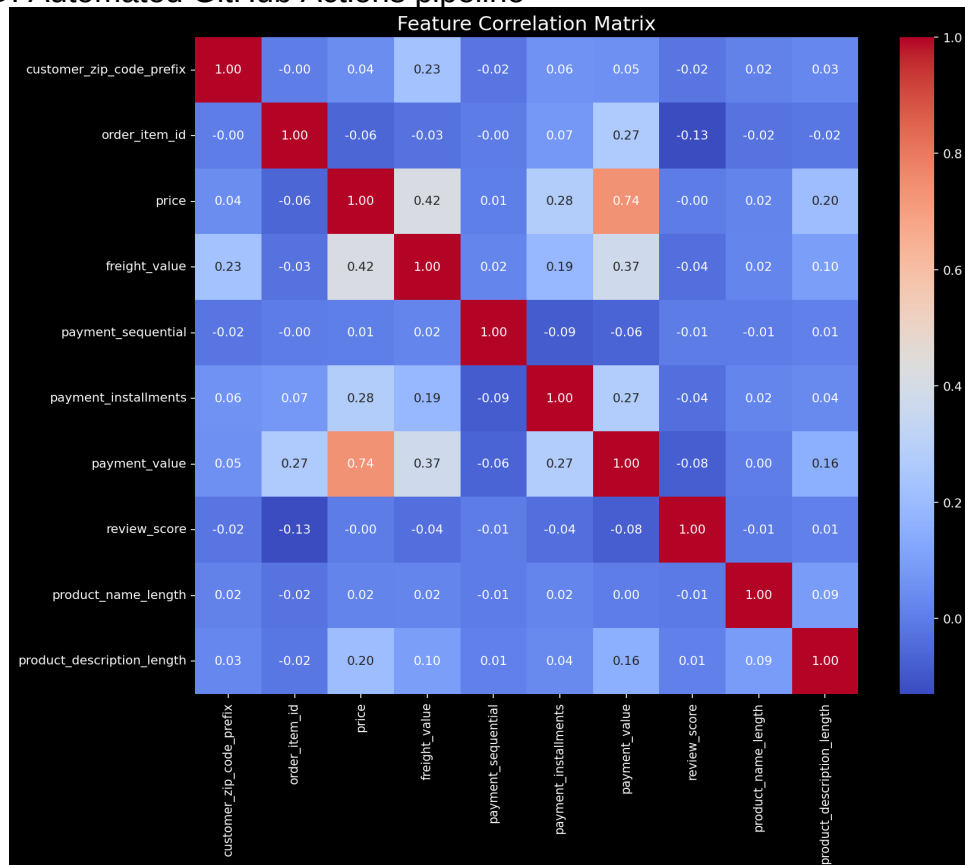


Figure 4: Feature correlation matrix used for dimensionality reduction.

7. Personal Reflection & Learnings

During this 4-week internship at Zidio Development, I have transitioned from building simple notebooks to deploying full-stack AI solutions. The most significant learning was the importance of MLOps—ensuring that a model is not just accurate on test data but stable and observable in production.

Key Learnings:

- Hybrid forecasting techniques for complex seasonality.
- Implementation of Drift Detection as a silent quality guardrail.
- Professional dashboard architecture with Streamlit.

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